



DYNO POWER 90

TECHNICAL DATASHEET

Dyno Power 90 is a high-strength, cap-sensitive packaged emulsion explosive with a firm, putty-like consistency. It is sensitized using chemical gassing, microspheres, or a combination of both, providing a stable and reliable energy output. The emulsion matrix offers excellent water resistance, allowing dependable performance in wet or partially flooded boreholes.

Dyno Power 90 is designed mainly for priming duties but can also be used as a column explosive in hard rock surface mining, underground operations, quarrying and tunneling. Its high detonation velocity and robust formulation make it highly effective for initiating ANFO columns and other non-cap-sensitive bulk or packaged explosives.

The cartridges are packed in durable, tear-resistant film that maintains shape during handling and ensures stability during water rise. This makes it suitable for demanding blasting conditions where consistent initiation and strong explosive energy are required.

TECHNICAL DETAILS

Density : 1.15 +/- 0.05 g/cc

Relative Weight Strength : 115%

Relative Bulk Strength : 172%

Velocity of Detonation : 4700 +/- 200 m/s

**RWS and RBS are calculated relative to ANFO at a density of 0.8 g/cc.

KEY BENEFITS

- High energy output with strong pre-compression resistance, ensuring consistent blasting performance in hard and confined ground conditions.
- Excellent water resistance, allowing reliable use in wet and water-logged boreholes without loss of energy.
- High velocity of detonation and strong weight strength for superior fragmentation and overall blast efficiency.
- Safe to manufacture, transport, store, and handle—classified as an oxidizer, reducing risk during non-blasting stages.
- Suitable for use as a primer, offering a cost-effective and safer alternative to cast/emulsion boosters.
- High viscosity and rigidity help maintain cartridge shape and ensure smooth loading even in varied hole conditions.
- Minimizes leaching, reducing environmental impact in sensitive areas.





SAFETY & STORAGE

- Store in a cool, dry, well-ventilated, licensed magazine approved for Class-2 explosives.
- Avoid exposure to high temperatures, direct sunlight, or freezing conditions, as extreme temperatures may reduce shelf life or alter performance.
- Maintain proper inventory rotation: always use the oldest stock first.
- Shelf life is generally 12 months under recommended storage conditions; deterioration may occur sooner if exposed to heat, moisture, or rough handling.
- Protect cartridges from shock, friction, or mechanical impact to prevent accidental initiation.
- Dispose of deteriorated or damaged explosives as per local regulations and under supervision of authorized personnel.

RECOMMENDATIONS FOR USE

- Suitable for blasting in dry, wet, and watery holes..
- Can be initiated using electric or non-electric detonators (minimum No. 8 strength) or detonating cord of 5 g/m PETN or higher, as per product specification.
- Achieve maximum energy output by proper tamping using a wooden or non-metallic tamping rod.
- Do NOT tamp a primer cartridge containing a detonator.
- Post-blast fumes are generally suitable for surface and underground applications, but ensure adequate ventilation before re-entry.
- Avoid excessive shock, heat, friction, or mechanical impact to prevent premature initiation.
- Suitable for ground temperatures in range -5°C to 50°C. For operations outside this range, seek guidance from the manufacturer or technical representative.

Transport
Classification

Class

UN
Number

Standard
Packaging

1.1D

UN 0241

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Diameter (mm)	Length (m)	Mass (g)	Units/box
25	215 +/- 5	125	200
32	215 +/- 5	200	125
40	310 +/- 5	400	63

